DUT: ATmega16U2

File: dut.kicad_sch

Pi to DUT: level translators, reset, HWB

File: pi_io.kicad_sch

Clock override (option CLK) and Si5351 (option SI)

File: clock.kicad_sch

DUT power switch and power meter

File: dut_power.kicad_sch

High-voltage programming (option HV)

File: hv.kicad_sch

USB path switch

File: usb.kicad_sch

Compute Module 4/5 connectors

File: cm.kicad_sch

Carrier: USB-C power and rpiboot, Ethernet, microSD, LEDs

File: carrier_io.kicad_sch

Generated by tools/schgen. Edit the generator, not this file.

TomSawyerLabs mcu-hil

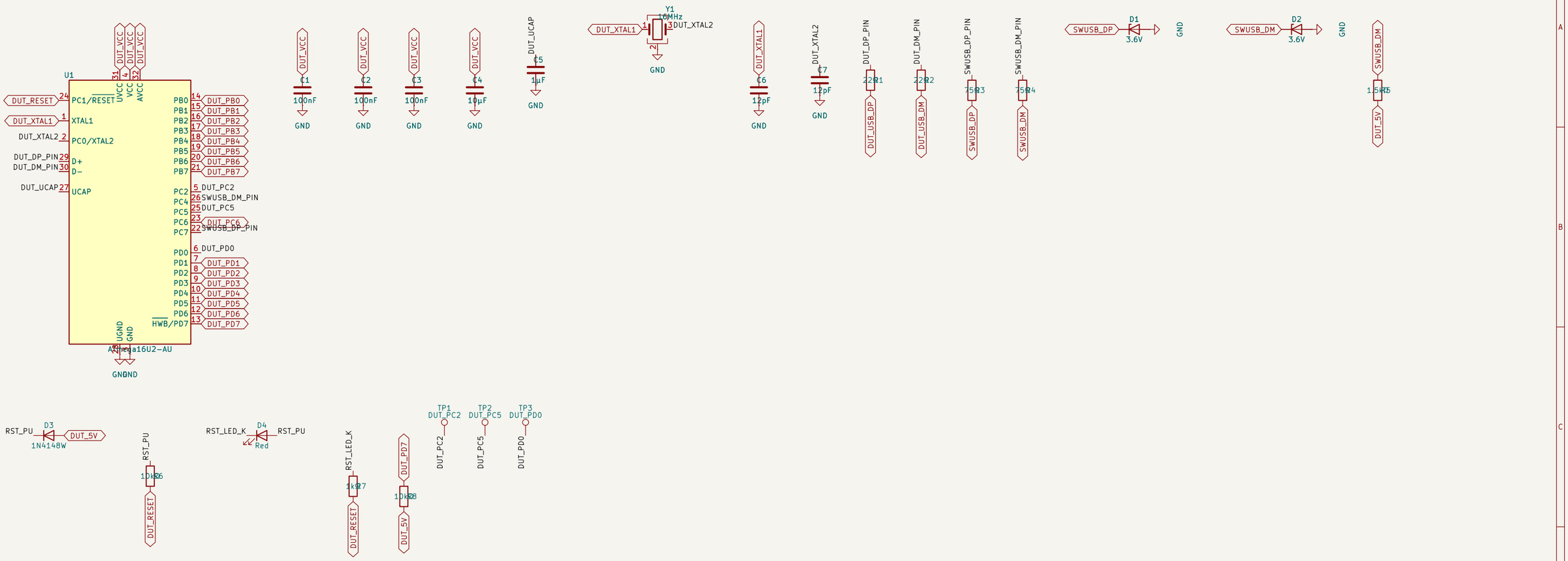
Sheet: /

File: mcu-hil-carrier.kicad_sch

Title: mcu-hil v2: CM4/CM5 carrier, ATmega16U2

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/9

ATmega16U2 at 5 V, 16 MHz. DUT_VCC is measured by the power meter; DUT_5V (before the shunt) feeds pull-ups.
Software USB (V-USB): D+ = PC7 (INT4), D− = PC4. 75R series and 3.6 V zeners keep the lines at USB levels.
Reset: the Pi side holds RESET low by default (pi_io sheet). The diode keeps 12 V (HVPP) off DUT_5V.



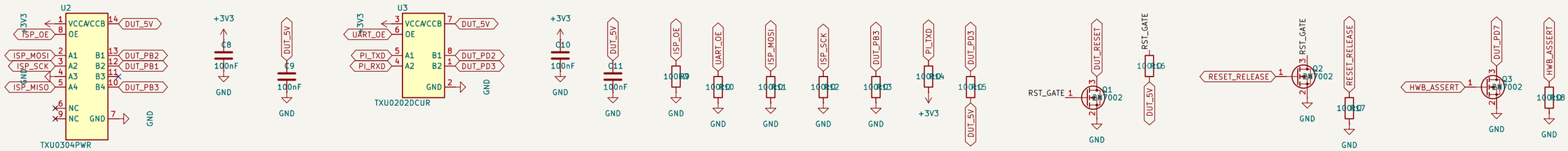
Generated by tools/schgen. Edit the generator, not this file.
TomSawyerLabs_mcu-hil

Sheet: /DUT: ATmega16U2/
File: dut.kicad_sch

Title: DUT: ATmega16U2

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/9

Translators: A side on +3V3 (Pi), B side on DUT_5V, so they go Hi-Z (off) whenever the DUT is off.
Every enable defaults off through an external pull-down (docs/v2-pinmap.md).
RESET: Q2 off (GPIO12 low, default) lets RST_GATE rise with DUT_5V, so Q1 holds RESET low.



Generated by tools/schgen. Edit the generator, not this file.

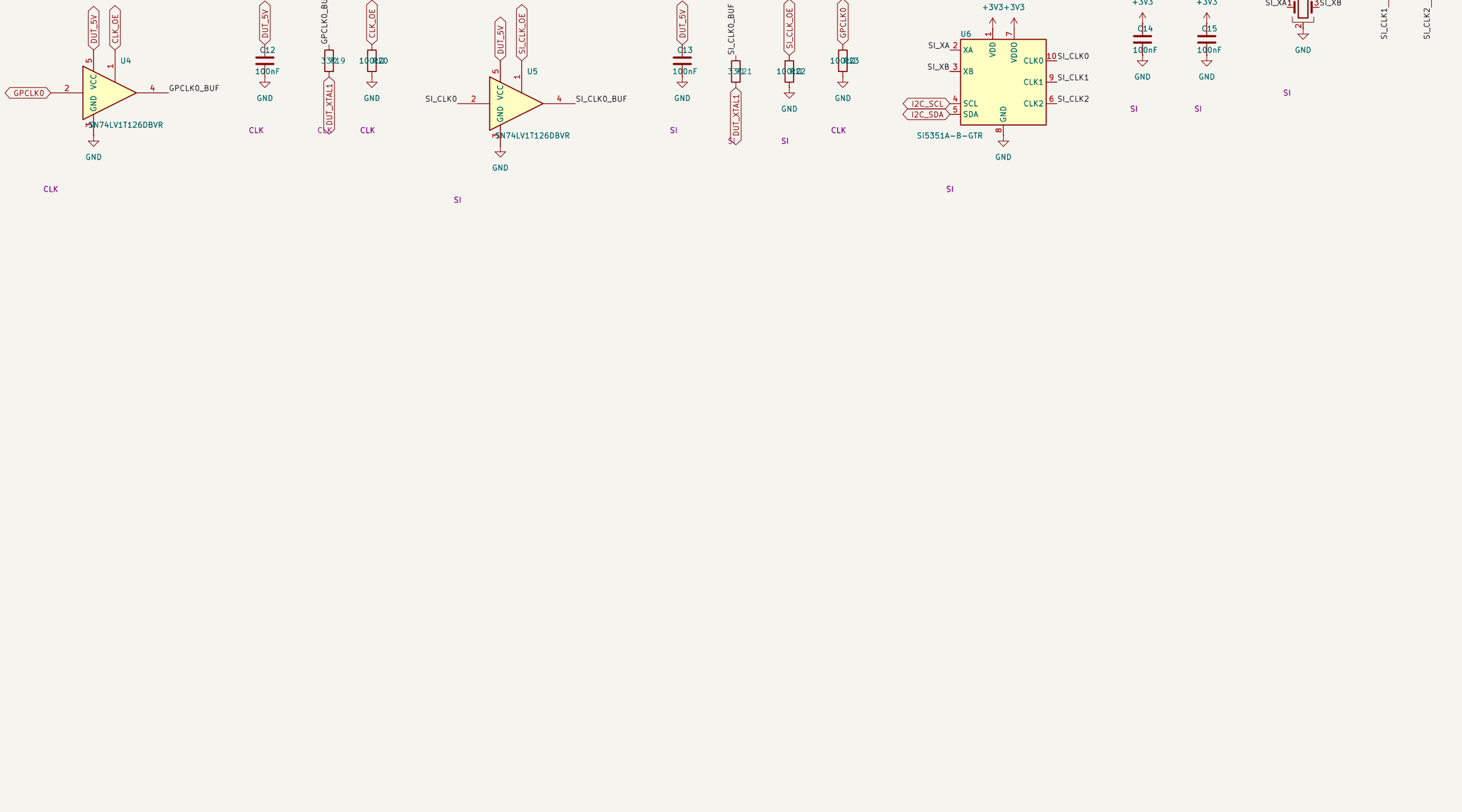
TomSawyerLabs mcu-hil

Sheet: /Pi to DUT: level translators, reset, HWB/
File: pi_io.kicad_sch

Title: Pi to DUT: level translators, reset, HWB

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/9

Two tri-state buffers drive XTAL1 over the crystal: GPCLK0 (GPIO4, enable GPIO18) or Si5351 CLK0 (enable GPIO27). Both default off. Software must never enable both. HVPP pulses XTAL1 through U(GPCLK0).



Generated by tools/schgen. Edit the generator, not this file.

TomSawyerLabs mcu-hil

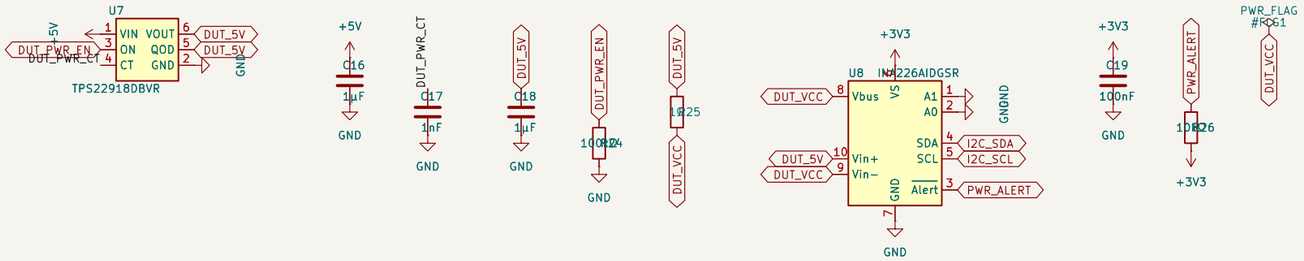
Sheet: /Clock override (option CLK) and Si5351 (option SI)/

File: clock.kicad_sch

Title: Clock override (option CLK) and Si5351 (option SI)

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/9

TPS22918: off by default (GPIO22 pull-down), set slew (CT), quick output discharge (QOD tied to VOUT).
INA226 measures DUT_VCC (the MCU only) across the 1 ohm shunt: 81.9 mA full scale, 2.5 uA steps.
INA228 (20-bit, same footprint and pinout) is a drop-in upgrade when JLC has stock: 164 mA, 0.3 uA steps.



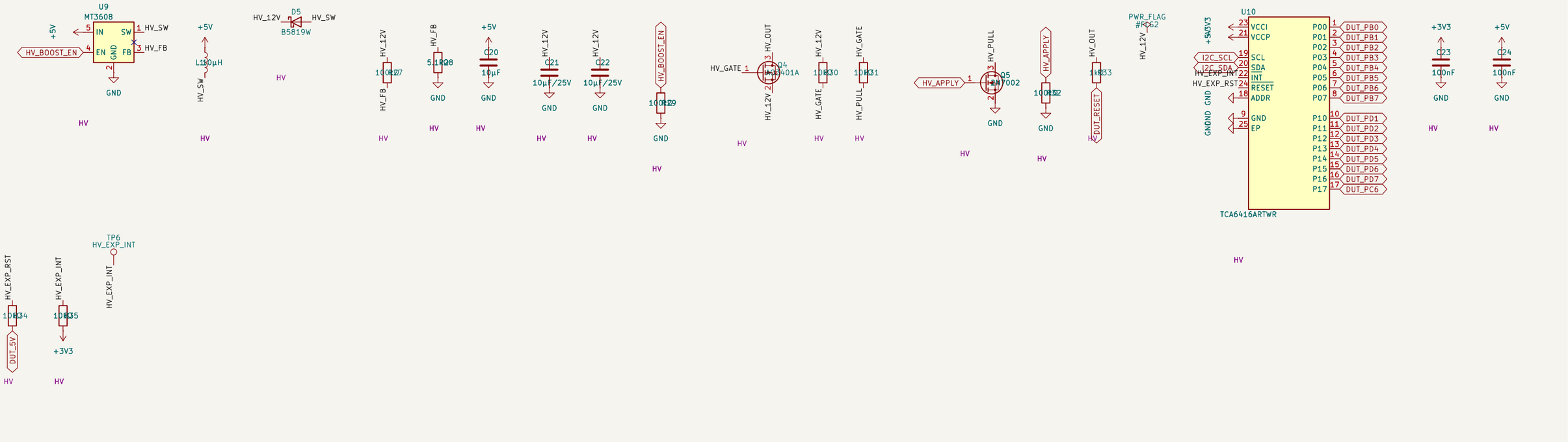
Generated by tools/schgen. Edit the generator, not this file.
TomSawyerLabs mcu-hil

Sheet: /DUT power switch and power meter/
File: dut_power.kicad_sch

Title: DUT power switch and power meter

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/9

12 V boost (MT3608; GPIO26) and a switch that puts it on RESET (GPIO25). Both default off.
FB divider 100k/5.1k: 12.36 V nominal. The 1k series resistor drops about 0.25 V at the 250 uA RESET current, so RESET sees 11.9–12.4 V (spec 11.5–12.5 V).
TCA6416A: VCCP on +5V (always up before +3V3, as TI recommends). ~RESET is pulled up to DUT_5V, so the expander's ports are held as inputs (Hi-Z) whenever the DUT is off. I2C address 0x20.
HVPP map (16U2 table 25-9): P0 = PB0-7 data; P10 PD1 RDY/BSY, P11 PD2 OE, P12 PD3 WR, P13 PD4 BS1, P14 PD5 XA0, P15 PD6 XA1, P16 PD7 PAGEL, P17 PC6 BS2.



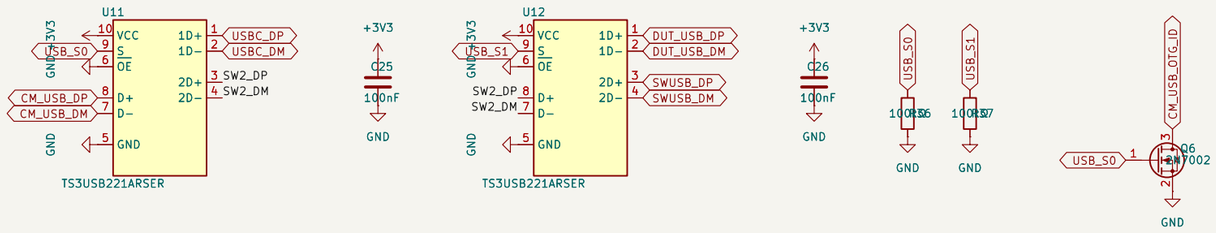
Generated by tools/schgen. Edit the generator, not this file.
TomSawyerLabs mcu-hil

Sheet: /High-voltage programming (option HV)/
File: hv.kicad_sch

Title: High-voltage programming (option HV)

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/9

USB_S0 (GPIO20) low: CM USB to the USB-C port (rpiboot; CM is a device). High: to the DUT side, and
Q pulls USB_OTG_ID low so the CM becomes the host. USB_S1 (GPIO21): low = 16U2 hardware USB, high = software USB.



Generated by tools/schgen. Edit the generator, not this file.

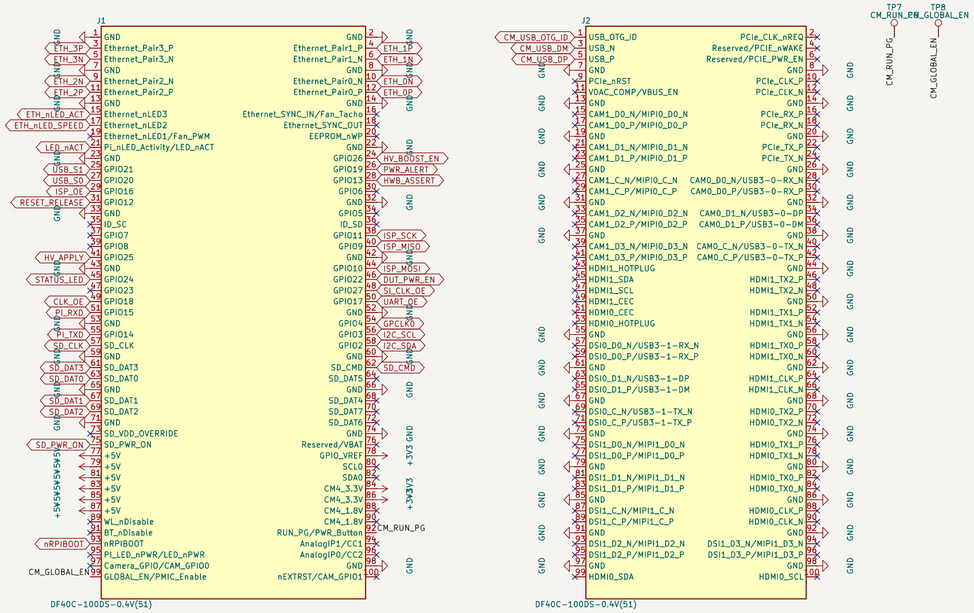
TomSawyerLabs mcu-hil

Sheet: /USB path switch/
File: usb.kicad_sch

Title: USB path switch

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 7/9

Hirose DF40C-100DS-0.4V(51). 1.5 mm stack. CM4 and CM5 share every pin used here.
GPIO_VREF to +3V3 (3.3 V GPIO). ID_SD/ID_SC, GLOBAL_EN/PMIC_EN, EEPROM_nWP and the 1.8 V outputs are unused.
CM5's USB-C CC pins (94/96) are left open: the carrier's own 5.1k Rd works for both CM4 and CM5
(CM5 then runs without 5 A PD negotiation).





Generated by tools/schgen. Edit the generator, not this file.

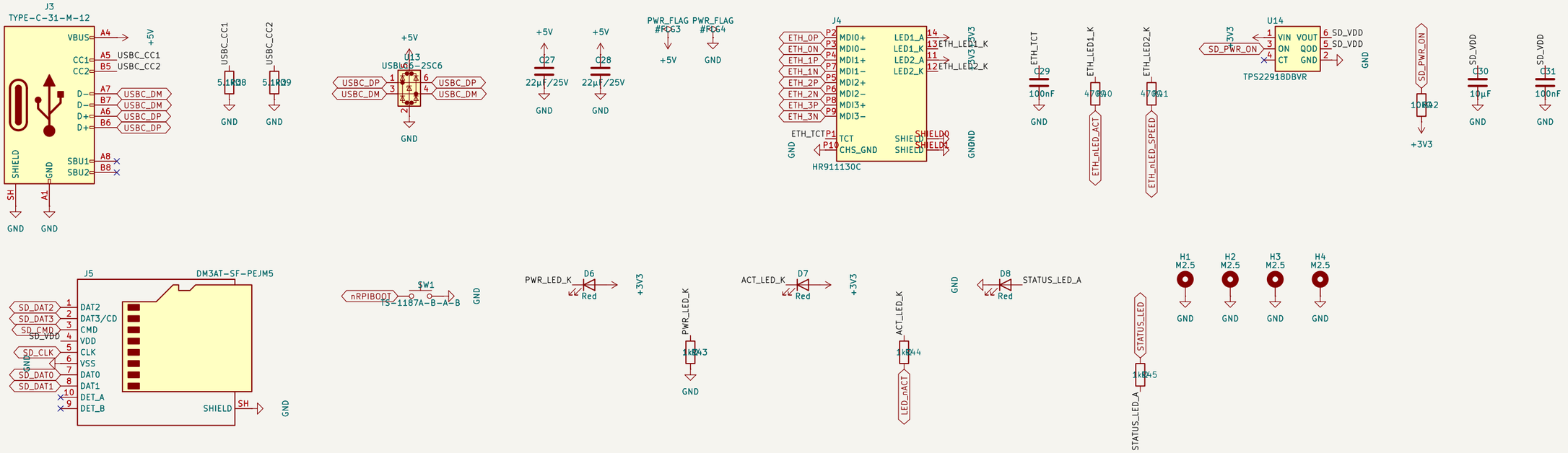
TomSawyerLabs mcu-hil

Sheet: /Compute Module 4/5 connectors/
File: cm.kicad_sch

Title: Compute Module 4/5 connectors

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/9

USB-C: 5 V input for the whole board and the CM's rpiboot/data port. Rd 5.1k on CC1/CC2 (sink).
Ethernet: 1:1 gigabit MagJack. PHY-side centre tap to GND through 100 nF; LEDs from +3V3 into the CM's open-drain nLED pins.
microSD (CM Lite): SD power through a second TPS22918 switched by SD_PWR_ON (CM4 needs the external pull-up).



Generated by tools/schgen. Edit the generator, not this file.

TomSawyerLabs mcu-hil

Sheet: /Carrier: USB-C power and rpiboot, Ethernet, microSD, LEDs/
File: carrier_io.kicad_sch

Title: Carrier: USB-C power and rpiboot, Ethernet, microSD, LEDs

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 9/9